Section A : Pure Mathematics [40 marks]

1 **(a)** On a single Argand diagram, shade the region satisfied by

$$|z-3-2i|\le 2 \quad \text{and} \quad \arg\left(\frac{z-2}{1+i}\right)\ge 0.$$ **[3]**

Find the range of $|z-5|$. **[2]**

(b) In an Argand diagram, points S and T represent the complex numbers $1+2i$ and $4-2i$. Without the use of a graphic calculator, find the angle *SOT*. **[3]**

2 In a controlled experiment, the rate at which liquid fuel is pumped into a tank is twice the remaining volume of the tank, and the rate at which the fuel in the tank is used up is proportional to the square of the volume of fuel in the tank. The capacity of the tank is $\frac{3}{2}$ m^3 and the volume of fuel in the tank at time t is x m^3.

(i) Given that the volume of fuel in the tank stabilizes at 1 m^3, show that the increase of the fuel is given by the differential equation $\frac{\mathrm{d}x}{\mathrm{d}t}=-\left(x^2+2x-3\right)$. **[2]**

(ii) Given that the tank is initially empty, find x in terms of t. **[6]**

(iii) Sketch the graph of your solution curve in **(ii)**. **[2]**

3 **(a)** The parametric equations of a curve are

$x = a(1-\cos t),\ y = a(t+\sin t)$, where a is a positive constant and $0 \le t \le 2\pi$.

(i) Sketch the curve. **[2]**

(ii) The region bounded by the curve and the y-axis is denoted by R. Find the volume of solid formed when R is rotated through 2π radians about the y-axis. **[4]**

(b) A curve has equation $y^2 = (1-x)^2 (x+3)$, where $-3 \le x \le 1$.

(i) Write down the integral that gives the area enclosed by the curve. **[1]**

(ii) By means of the substitution $u^2 = x+3$, find the area enclosed by the curve. **[4]**

4 The equations of three planes $p_1,\ p_2,\ p_3$ are

$$5x+3y-z=2,$$

$$x-2y+3z=3,$$

$$\alpha x - y + 2z = \beta,$$

respectively, where α and β are constants.

The planes p_1 and p_2 intersect in a line l.

(i) Find a vector equation of l. **[2]**

(ii) If the planes p_1 and p_4 are parallel and the point (1, 1, 2) is equidistant from these two planes, find the equation of the plane p_4 in scalar product form. **[3]**

(iii) Given that the planes p_2 and p_3 intersect at right angles, find α . **[2]**

(iv) Given that the point $(1,-1,0)$ lies on all three planes, find a possible set of values of α and β. **[2]**

(v) Find the geometrical relationship of the three planes when $\alpha = \frac{10}{7}$ and $\beta = \frac{17}{7}$.

Deduce the geometrical relationship of the three planes when $\alpha = \frac{10}{7}$ and $\beta \ne \frac{17}{7}$. **[2]**

Section B : Statistics [60 marks]

5 **(a)** Explain why it is advisable to plot a scatter diagram before interpreting a correlation coefficient calculated for a sample drawn from a bivariate distribution. **[1]**

(b) The table below shows the age and average head circumference of 11 baby boys.

Age (month) x	Head circumference (cm) y
1	40.7
2	42.3
3	43.6
6	45.9
9	47.4
15	48.5
18	49.3
21	50
24	51
33	52
36	52.2

(i) Calculate the linear product moment coefficient between x and y. **[1]**

Give a sketch of the scatter diagram for the data of the eleven baby boys and hence comment on the suitability of finding the linear regression line y on x. **[3]**

(ii) State with a reason, which of the following would be an appropriate model to represent the above data.

A: $y = a + bx^2$

B: $y = a + \dfrac{b}{x}$

C: $\ln y = a + b \ln x$

For the most appropriate model, determine the least square estimates of a and b. **[3]**

6 The height of girls, chosen at random, is found to have a nominal mean of 155cm with a standard deviation of 10cm. Find

(i) the range of the girls' height if 40% of the girls have heights falling symmetrically within the mean, **[3]**

(ii) the probability that exactly two girls are taller than 170cm if four girls are chosen, **[3]**

(iii) the probability that the total height of n randomly chosen girls is more than $150n$ cm, if n is large. **[3]**

7 **(a)** A teacher claims that the average time a student spends on mathematics revision per day is 120 minutes. A random sample of 150 students is observed and the time, x min, that each student spends on mathematics revision per day noted. It is found that

$$\sum(x-120) = -1500 \quad \text{and} \quad \sum(x-120)^2 = 462\ 000 .$$

(i) Test, at the 1% level of significance, whether the teacher has overstated the average time spent on mathematics revision per day by a student. **[4]**

(ii) What is the least number of students that is required in the sample to arrive at a different conclusion from that in part **(i)**? **[2]**

(b) The manufacturer of substance Y claims that the average iron content in substance Y is not more than 4. Eight samples of substance Y are selected and the iron content in each sample is tested and recorded as follows:

4.07, 3.98, 4.01, 4.2, 4.03, 4.08, 3.9, 4.44

(i) Using an appropriate test, determine the largest level of significance for which the null hypothesis is rejected. **[3]**

(ii) State an assumption necessary for the test in **(i)** to be valid. **[1]**

8 On average, the number of cases of dengue fever in Country X is 1 in 100,000 inhabitants per month. Country X consists of only City A and City B and each city has a population of 250,000 and 400,000 respectively such that the number of cases of dengue fever for both cities is 1 in 100,000 inhabitants per month. The inhabitants from both cities infected with dengue fever are independent of one another.

Find, **[1]**

(i) the probability that City A have more than 3 cases of dengue fever in a month,

(ii) the probability that number of cases of dengue fever in a month in City A is less than 2 given the probability that the total number of cases in a month in City A and City B is 4, **[3]** **[4]**

(iii) using a suitable approximation, the probability of both cities having a total of less than 15 cases of dengue fever in 2 months, **[2]**

(iv) the probability that City A will have at least 8 months during a particular year that have at most 3 cases of dengue fever in a month. **[1]**

Give a reason why the answer that you have found in part **(iv)** is not valid.

9 **(a)** Patients who caught the common flu have a 98% chance of full recovery. Find the probability that

(i) in a sample of 20 patients, more than 2 will not recover fully, **[2]**

(ii) in fifty sample of 20 patients, the mean number of patients who will not recover is more than 0.5. **[3]**

(b) At the onset of a particular contagious flu virus, the probability that a patient survives is given by p. In a hospital ward of 50 patients, pass data shows that the probability of more than 48 patients surviving the flu is 0.64.

(i) Find the probability that a patient will survive the flu. **[2]**

(ii) Using a suitable approximation, find the probability that less than 46 patients survive in a group of 50 patients. **[4]**

10 **(a)** How many three-letter codeword can be formed from the word "DISTINCTION"? **[3]**

(b) A bag contains 10 red marbles and 90 blue marbles which are indistinguishable, apart from colour. Marbles are drawn from the bag singly and at random, with replacement.

(i) Calculate the probability that the first red marble is obtained on or before the 5^{th} draw. **[2]**

(ii) Given that no red marbles is obtained after 5 draws, find the probability that the first red marble is obtained on the 8^{th} draw. **[3]**

(iii) Shawn and Matthias took turns to draw marbles from the above mentioned bag. The colour is observed and the marble is replaced. The first person who draws a red marble wins the game and the game will stop. Given that Matthias starts first, find the probability that Shawn wins the game. **[3]**

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